

How to View This Project

Trading Against Noise · SHBI-GB.7301 Final Project · Larionova, Shen, Xi, Zheng

1. The report

Open “Final Report.pdf” in this folder (19 pages). The editable Word version and everything that produced it live in the background folder (background/report/Stochastic_Modeling_Final_Report.docx).

2. The slides

Double-click “Trading Against Noise — Slides.html”. It opens in any web browser with no setup: use the left/right arrow keys (or click) to move through the 19 slides, and press N to show the speaker notes on each slide.

3. Running the simulation and interactive dashboard (optional)

The only requirement is Python 3.11 or newer (free from python.org). One command sets up an isolated environment, installs the packages, runs the 37-test validation suite, and opens the interactive dashboard in your browser:

- Windows: open the background folder and double-click run.bat
- Mac (or Linux): open Terminal, then run:

```
cd path/to/background      (drag the folder into Terminal to fill the path)

python3 run.py
```

The first run takes a few minutes to install packages; after that it starts in seconds. The dashboard appears at <http://localhost:8601> and works fully offline (five real Polymarket price histories are bundled). Useful variants:

- `python run.py --experiments` regenerates every result CSV and figure from the master seed (20260812)
- `python run.py --slides` serves the slide deck with a demo button that boots the dashboard in one click

Where everything lives

- background/src/ – the simulation engine, trading policies, experiments, calibration, and validation code
- background/tests/ – the 37-test validation suite (run with: `python -m pytest`)
- background/results/ – every experiment CSV, the numeric summary, and the review log;
background/report/figures/ – all figures
- background/notebooks/analysis.ipynb – the executed analysis walkthrough

Everything is also on GitHub: github.com/larionovaa2248-glitch/stochastic-project